INCH-POUND

MIL-DTL-32341A (MR)
w/AMENDMENT 2
5 SEPTEMBER 2024
SUPERSEDING
MIL-DTL-32341A (MR)
w/AMENDMENT 1
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DETAIL SPECIFICATION

ARMOR PLATE, ALUMINUM, ALLOY 2139 WELDABLE and ALLOY 2195 and 2060 UNWELDABLE APPLIQUE

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers three wrought aluminum armor plate alloys for both welded and un-welded applications in nominal thicknesses from 0.500 to 4.000 inch, inclusive (see 6.2). An additional application for the AA2139 alloy with the T84 temper is also included in this specification for ballistic plate with an ordered thickness of 1.500 to 2.600 inches. The weldability of wrought aluminum alloy AA2139 armor has been determined for class I, type A armor for these thicknesses, only; 0.500" and 1.500". The weldability for class I, type B and the other alloys (class II & class III) has not been determined and therefore should only be used as appliqué armor.

1.2 Weldability. Class I material covered by this specification has been demonstrated to be weldable to itself and some other weldable alloys (see 6.3 and 6.4).

1.3 Classification. The wrought aluminum armor should be of the following classes, as specified (see 6.2).

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate, Specifications and Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 9535

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1.3.1 Class I. Class I is wrought aluminum armor that conforms to the Aluminum Association designation for the 2139 aluminum alloy.

1.3.1.1 Type A. Type A has a temper designation (See 6.6) which is a high-strength temper designed for maximum resistance to armor piercing (AP) projectiles. The applicable gauge range for class I, type A is 0.500-4.000 inches.

1.3.1.2 Type B. Type B has a temper designation (See 6.6) which is designed for maximum resistance to blast. The applicable gauge range for class I, type B is 1.501 to 2.600 inches.

1.3.2 Class II. Class II is wrought aluminum armor that conforms to the Aluminum Association designation for the 2195 aluminum alloy. The applicable gauge range for class II is 0.500 - 2.250 inches.

1.3.3 Class III. Class III is wrought aluminum armor that conforms to the Aluminum Association designation for the 2060 aluminum alloy. The applicable gauge range for class III is 0.500 – 2.500 inches.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662	-	V_{50} Ballistic Test for Armor
MIL-STD-3057	-	Arc Welding of Armor Grade Aluminum

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal should apply.

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THE ALUMINUM ASSOCIATION, INC.

ANSI H35.2	-	American National Standard Dimensional Tolerances for Aluminum Mill Products
Teal Sheets	-	International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys

(Copies of these documents are available online at https://www.aluminum.org/.)

ASTM INTERNATIONAL

ASTM B557	-	Standard Test Methods for Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products
ASTM E716	-	Standard Practices for Sampling Aluminum and Aluminum Alloys for Spectrochemical Analysis
ASTM E1251	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Atomic Emission Spectrometry
ASTM E3061	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Inductively Coupled Plasma Atomic Emission Spectrometry (Performance Based Method)
ASTM G38	-	Standard Practice for Making and Using C-Ring Stress-Corrosion Test Specimens
ASTM G47	-	Standard Test Method for Determining Susceptibility to Stress-Corrosion Cracking of 2XXX and 7XXX Aluminum Alloy Products

(Copies of these documents are available online at https://www.astm.org/.)

SAE INTERNATIONAL

SAE AMS 2750	-	Pyrometry

(Copies of these documents are available online at https://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), first article testing shall be required and all test samples required by this specification shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. First article testing shall be completed before production material is submitted for acceptance testing. The approval of the first article samples authorizes commencement of production but does not relieve the supplier of the responsibility to comply with all the applicable provisions of this specification. The first article samples and acceptance test plates shall be manufactured by the process proposed for use on production items.

3.2 Chemical composition. The chemical compositions for classes I, II, and III alloys defined by the Aluminum Association Teal Sheets, "International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys" are listed within table I and shall be within those limits. A certification of conformance of the chemical composition of the alloy shall be furnished with the ballistic test plates.

TABLE I. Chemical composition, weight percent. 1/

ELEMENTS	SYMBOL	Class I 2139 ALLOY 2/	Class II 2195 ALLOY 2/	Class III 2060 ALLOY 2/
Silicon	Si	0.10	0.12	0.07
Iron	Fe	0.15	0.15	0.07
Copper	Cu	4.5 -5.5	3.7 - 4.3	3.4 - 4.5
Manganese	Mn	0.20 -0.60	0.25	0.10 - 0.50
Magnesium	Mg	0.20 - 0.80	0.25 - 0.8	0.60 - 1.1
Chromium	Cr	0.05	N/A	N/A
Zinc	Zn	0.25	0.25	0.30 - 0.50
Titanium	Ti	0.15	0.10	0.10
Vanadium	V	0.05	N/A	N/A
Zirconium	Zr	N/A	0.08 - 0.16	0.05 - 0.15
Lithium	Li	N/A	0.8 - 1.2	0.60 - 0.90
Silver	Ag	0.15 - 0.60	0.25 - 0.6	0.05 - 0.50
Other, max. Each	---	0.05	0.05	0.05
Other, max. Total 3/	---	0.15	0.15	0.15
Aluminum	Al	Remainder	Remainder	Remainder

1/ Except for "Aluminum" and "others", analysis normally is made for elements for which specific limits are shown.
2/ Where single units are shown, these indicate the maximum amounts permitted.
3/ The sum of those "others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum.

3.3 Mechanical properties. Unless otherwise specified in the contract or purchase order (see 6.2), the mechanical properties of the test specimen taken in the longitudinal (L) direction shall meet the minimum mechanical properties listed in table IIa for class I or in table IIb for classes II and III. If mechanical property requirements differ from those contained in tables IIa or IIb, or if any

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other properties are required, the ballistic requirements shall be negotiated between the procuring activity and the supplier.

TABLE IIa. Minimum mechanical properties. 1/ 2/

Thickness, inches	Tensile Strength, ksi		Yield Strength, 0.2% Offset, ksi		Elongation percent	
	Class I Type A	Class I Type B	Class I Type A	Class I Type B	Class I Type A	Class I Type B
0.500 to 3.000, incl.	67	62 3/	64	53 3/	9	10 3/
3.001 to 4.000, incl.	67	N/A	64	N/A	9	N/A

1/ The test specimen gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch; all other thicknesses shall have a gage length of 2.000 inches.
2/ Values are taken in the Longitudinal Direction (see 3.3).
3/ For alloy 2139 (class I, type B) the applicable thickness range is 1.501-2.600 inches.

TABLE IIb. Minimum mechanical properties. 1/ 2/

Thickness, inches	Tensile Strength, ksi		Yield Strength, 0.2% Offset, ksi		Elongation percent	
	Class II	Class III	Class II	Class III	Class II	Class III
0.500 to 2.500, incl.	71 3/	73	63 3/	70	9 3/	7

1/ The test specimen gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch; all other thicknesses shall have a gage length of 2.000 inches.
2/ Values are taken in the Longitudinal Direction (see 3.3).
3/ For alloy 2195 (class II) the applicable thickness range is 0.500-2.250 inches.

3.4 Weldability. When the vehicle requires weldments the fabricators (OEMs and kit manufacturers, etc.) are required to demonstrate weldability (which includes the Proofing Projectile Impact Testing (PPIT) formerly known as ballistic shock testing) for armor to armor and armor to non-armor applications in accordance with MIL-STD-3057, "Arc Welding of Armor Grade Aluminum". Since the alloys referenced by this specification are not all listed in MIL-STD-3057, it is strongly suggested that the contract or drawing for the vehicle specify weldability requirements as outlined in paragraph 6.3. PPIT procedures and associated striking velocities for a variety of class I and class III plate thicknesses are listed in MIL-STD-3057. For any newly introduced armor material, configuration, or unproven thickness being used for armor applications, the aluminum fabricators shall qualify a weld procedure in accordance with MIL-STD-3057, and this requires PPIT testing.

3.5 Ballistic limit. The protection ballistic limit, BL(P), shall be as specified in Appendix A. When a complete penetration cannot be obtained for any class of armor material, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic

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velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

3.6 Stress corrosion resistance. When stress corrosion testing is specified in the contract or purchase order (see 6.2) and unless otherwise specified in the contract or purchase order (see 6.2), plate 0.750 inch and over in thickness shall be resistant to stress corrosion cracking, with a minimum of 5 of the 9 specimens showing no evidence of cracking at the end of 96 hours.

3.7 Thermal processing. Heat treatment shall conform to the requirements of SAE AMS 2750 and shall be such as to enable the material to meet the requirements of these specifications.

3.8 Dimensions. Dimensions for plates delivered for fabrication shall have an overall dimension tolerance of +0.500/-0.000 for width and length unless otherwise specified in the contract or purchase order (see 6.2).

3.8.1 Tolerances. Unless otherwise specified in the contract or purchase order (see 6.2), the plates shall not vary from the specified ordered dimensions by an amount greater than that specified by ANSI H35.2, except for thickness.

3.8.2 Thickness. Thickness tolerance for production armor and ballistic test plates shall be as specified in table III.

TABLE III. Thickness tolerances.

Ordered Thickness (Inches)		SPECIFIED WIDTH (Inches)								
		OVER	0.00	39.37	59.06	78.74	98.43	118.11	137.80	157.48
		THRU	39.37	59.06	78.74	98.43	118.11	137.80	157.48	177.17
OVER	THRU	TOLERANCES - INCHES (PLUS and MINUS)								
0.500	1.000		0.031	0.031	0.037	0.043	0.051	0.060	0.070	0.085
1.000	1.575		0.039	0.039	0.047	0.055	0.065	0.075	0.090	0.105
1.575	2.362		0.055	0.055	0.060	0.070	0.085	0.100	0.115	---
2.362	3.000		0.075	0.075	0.085	0.100	0.105	0.125	---	---
3.000	4.000		0.090	0.090	0.105	0.110	0.125	---	---	---

3.9 Marking for identification. Unless otherwise specified in the contract or purchase order (see 6.2) each plate shall be marked on one plate edge with the manufacturer's name or CAGE code, the basic number of this specification, the plate thickness in inches, the alloy designation and the lot number or code relating to the lot number (see 4.2). The height of the characters shall be 3/8 of an inch or greater. Impression stamping shall not be used unless permitted by the procuring activity (see 6.2). Each plate shall be marked in lengthwise rows of characters recurring at intervals not greater than 3 feet, the rows being spaced not more than 6 inches apart and alternately staggered. The characters shall be not less than 3/8 inch in height and shall be applied using a suitable marking fluid whose residue shall not contain more than traces of halogen-bearing compounds and shall be capable of being removed in hot alkaline cleaning solution without rubbing. The markings shall have no deleterious effect on the plate material or its performance and shall be sufficiently stable to withstand normal handling.

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3.9.1 Ballistic test plates. In addition to the markings in 3.8, each ballistic test plate shall be marked with the letters PRE for First Article test plates and ACC for Acceptance test plates. This marking shall be impression stamped on the edge in letters 3/8 inch high or greater. Ballistic retest plates shall be marked "R1" and "R2" respectively (see A.5.2).

3.10 Ballistic test plate information. For each lot of aluminum alloy armor a properly completed Aluminum Armor Test Data Form (See figure 1) shall be submitted with each ballistic test plate that represents that particular processing lot.

3.11 Workmanship. Plate produced under this specification shall be uniform in quality and clean, smooth and sufficiently free from buckles, blisters, hard spots, damaged ends, laminations and other defects which may affect its use.

3.11.1 Surface and Edge Condition. The surface and edge condition of the plate shall be free of surface cracks, edge cracks, and edge laminations as defined in the contract or purchase order (see 6.2).

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REQUEST FOR BALLISTIC TEST OF ALUMINUM ALLOY ARMOR										
SPECIFICATION: MIL-DTL-32341			**REVISION: A**			**AMENDMENT: 2**				
FIRING RECORD:		**DATE:**			**TEST CONTRACT NO:**					
	PLATE MANUFACTURER / PRODUCER				**PRIME CONTRACTOR**					
Name:										
Address:										
POC:										
Phone No:										
Email:										
TEST ITEM IDENTIFICATION:					Ordered Thickness:					
Lot No._____ **Plate No.**_____			**Alloy:**_____ **Temper** _____			**Class:** _____ **Type:** _____				
PURPOSE:___ Acceptance ___ First Article ___ Development ___ Reduced Testing ___ Audit Testing										
SAMPLE: ___ Primary ___ Retest (Firing Record No. of Failed Sample _____)										
CHEMISTRY ANALYSIS	**Si**	**Fe**	**Cu**	**Mn**	**Mg**	**Cr**	**Zn**	**Ti**	**Zr**	
V	**Li**	**Ag**	**Ni**	**Mn+Cr**	**Zr+Ti**	**Other - Each**			**Other - Total**	**Al**
										Rem.
MECHANICAL PROPERTIES:										
UTS (ksi):			0.2% YS (ksi):			Elongation (%):				
WELDABILITY:	Thickness		Required Velocity		Actual Velocity		Pass/Fail			
BALLISTIC TEST RESULTS: **SEPARATELY HEAT TREATED TEST PLATE : YES ____ NO ____**										
	Obl. (deg)	**Actual Thickness (in)**	**Required V_{50} (fps)**	**Actual V_{50} (fps)**	**Pass/ Fail**					
Test:										
Projectile:										
	NOTES:									
SPALL:	Spall Size:		Two Projectile Calibers:		Pass / Fail:					
DO NOT FILL BELOW THIS LINE - TO BE COMPLETED BY GOVERNMENT OFFICIAL ONLY										
LOT: [____ met] [____ failed to meet] the ballistic requirements of specification MIL-DTL-32341										
Government Representative:		Date:	Supplier Representative:		Date:					

FIGURE 1. Aluminum Armor Test Data Form.

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Lot. A lot shall consist of all plates of the same alloy; same source of molten metal representing the ingots poured, and ordered thickness which has been processed together by the same mill practice. Unless otherwise specified in the contract or purchase order (see 6.2), the weight of the finished plate in the lot shall not exceed 50,000 pounds and shall be submitted for inspection as a unit. All lots shall be fully traceable back to the ingot source by the supplier.

4.3 First article inspection. When specified in the contract or purchase order, (see 6.2) and before production has commenced, samples of specified material shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples shall be produced using the mill production practice proposed for use on production.

4.3.1 First article inspection. First article inspection except as otherwise indicated in this specification, shall utilize the same requirements and test methods as the conformance inspection or production acceptance inspection shown in 4.4.

4.4 Conformance inspection. Conformance inspection or production acceptance inspection shall include the examination of 4.6 and the tests of 4.7.

4.5 Sampling.

4.5.1 First article inspection.

4.5.1.1 Chemical composition. One (1) sample for chemical analysis shall be removed from each plate selected for ballistic testing and shall meet the requirements of 3.2 when tested as specified in 4.7.1.

4.5.1.2 Mechanical properties. One tension test specimen shall be removed from each plate that has been selected for ballistic testing and shall meet the requirements when tested as specified in 4.7.2.

4.5.1.3 Ballistic tests. Two plates, 12 inches by 36 inches of each thickness to be supplied on the contract, shall be submitted for ballistic testing in accordance with Appendix A. The orientation of these plates with respect to the rolling direction shall be at the option of the producer (see 6.2).

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4.5.2 Conformance inspection.

4.5.2.1 Chemical composition. The sample shall meet the chemical composition requirements of 3.2 when tested as specified in 4.7.1.

4.5.2.1.1 Ingot analysis. At least one sample shall be taken from the molten metal representing the ingots poured as a unit from the same source molten metal. Complete ingot analysis records shall be available to the Government at the producer's facility.

4.5.2.1.2 Product analysis. When sampling has not been made in accordance with 4.5.2.1.1, one sample shall be randomly taken for each 4,000 pounds or less in a lot. Complete product analysis records shall be available to the Government at the contractor's facility.

4.5.2.2 Mechanical properties. Samples for tension tests shall be selected from each lot in accordance with table IV. Each sample shall be randomly selected from a different plate in the lot, and only one tension test specimen shall be made from each sample. The sample shall meet the requirements when tested as specified in 4.7.2.

TABLE IV. Number of tension tests.

Lot size, pounds	Minimum number of samples 1/
To 8,000, incl.	2
8,001 to 12,000, incl.	3
12,001 to 20,000, incl.	4
20,001 up	5

1/ If a lot consists of only one plate, one sample shall be required.

4.5.2.3 Ballistic testing. One plate, 12 inches by 36 inches, shall be randomly selected from each lot for ballistic testing. The orientation of the plate with respect to the rolling direction shall be at the option of the producer (see 6.2). The sample shall meet the requirements when tested as specified in 4.7.4.

4.5.2.4 Stress corrosion tests. When required (see 6.2), one plate, 12 inches by 12 inches, shall be randomly selected from each lot for stress corrosion testing. The plate shall meet the requirements when tested as specified in 4.7.5.

4.6 Examination.

4.6.1 Visual. Each plate shall be examined for compliance with the identification marking (see 3.8) and workmanship (see 3.10) requirements.

4.6.2 Dimensions. Plates within a lot shall be measured to determine compliance with requirements of paragraph 3.7 in accordance with the sampling procedures approved by the procuring activity and as specified in the contract or purchase order (see 6.2). At a minimum, one plate per lot shall be randomly selected for dimensional inspection.

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4.7 Test specimens.

4.7.1 Chemical composition. Samples for chemical analysis shall be prepared and tested in accordance with one or more ASTM methods of E716, E1251, and E3061. In case of dispute, analysis by method E3061shall be the basis for acceptance or rejection.

4.7.2 Mechanical properties. Tension test specimens shall be prepared and tested in accordance with ASTM B557. Specimens shall be taken in the longitudinal (L) direction. For plate less than 0.500 inch in thickness, a standard rectangular tension test specimen shall be used. For plate in nominal thickness 0.500 to 1.500 inches, inclusive, tension test specimens shall be taken with the axis midway between the two plate surfaces. For plate in nominal thickness greater than 1.500 inches, the axis of the tension test specimen shall be three-fourths of the distance from one surface to the other.

4.7.3 Weldability. When required (see 6.2), the thickness of the weldments specified in the contract or drawing shall be used to determine the proofing projectile and striking velocity in accordance with MIL-STD-3057.

4.7.4 Ballistic testing. The ordered thickness specified in the contract or purchase order shall be used to determine the test projectile in accordance with table V. Ballistic testing shall be in accordance with Appendix A. Test plate thickness, as measured by the ballistic testing agency, shall be used in conjunction with table V and Appendix A to determine the required V_{50} protection ballistic limit for that plate. Thickness shall be determined as the average of at least four thickness measurements read on a deep throat micrometer or by means of an ultrasonic device to the nearest 0.001 of an inch and rounded off to the nearest 0.005 of an inch. Measurements shall be made on the intended impact area. In those cases where the BL(P) is within ±10 fps of the minimum required value for the measured average thickness (to the nearest 0.005-inch), an interpolation of the appropriate ballistic limit table shall be performed. The average plate thickness, computed to the nearest 0.001-inch, shall be used to determine the minimum required BL(P) for that plate.

TABLE V. Acceptance ballistic test plates.

Ordered Thickness, Inches	Projectile	Angle of Obliquity in Degrees	TABLE	CLASS I (2139)	CLASS II (2195)	CLASS III (2060)
0.500 - 0.749	Cal. .30 AP M2	30	A-I	TBD [1/]	LISTED	LISTED
0.750 - 0.950	Cal. .50 FSP	0	A-II	LISTED - SAME V_{50}		LISTED
0.951 - 1.500	20-mm FSP	0	A-III	LISTED - SAME V_{50}		
0.750 - 1.500	Cal. .30 AP M2	0	A-IV	LISTED - SAME V_{50}		LISTED
1.501 - 2.600	Cal. .50 AP M2	0	A-V	LISTED [2/]	LISTED [3/]	LISTED [4/]
2.501 - 3.500	14.5-mm BS41	0	A-VI	LISTED	N/A [3/]	N/A [4/]
3.501 – 4.000	20-mm M602	0	A-VII	TBD	N/A [3/]	N/A [4/]

[1/] To be determined.
[2/] Class I, type A (alloy 2139) the applicable gauge for ordered thickness ends at 2.500 inches.
[3/] Class II (alloy 2195) the applicable gauge for ordered thickness ends at 2.250 inches.
[4/] Class III (alloy 2060) the applicable gauge for ordered thickness ends at 2.500 inches.

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4.7.4.1 Ballistic testing facility. Unless otherwise specified in the contract or purchase order (see 6.2), the ballistic test plates shall be forwarded to the Commander, USA ATC, ATTN: TEDT-AT-SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059 or to an approved Government facility for ballistic testing for first article or production acceptance.

4.7.4.2 Incomplete penetrations. When a complete penetration cannot be obtained, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

4.7.5 Capability requirements for resistance to stress corrosion. When stress corrosion testing is required (see 6.2), specimens shall be prepared in accordance with ASTM G38 and tested in accordance with ASTM G47 with nine specimens each prepared per sample per lot. The stress corrosion specimens shall be stressed to 45 ksi.

4.8 Rejection and retest. Unless otherwise specified in the contract or purchase order (see 6.2) and except as specified in 4.7.2 and 4.7.3, rejection and retest shall be conducted in accordance with 4.8.1, 4.8.1.1, and 4.8.2.

4.8.1 Rejection of first article plates. When one or more first article test specimens fail to meet the requirements of 4.3, the product lot and process, represented by the test plates or specimens shall be subject to rejection except as otherwise provided in a sampling plan approved by the procuring activity and in requirements of 4.8.1.1.

4.8.1.1 Retest of first article samples. Resubmission and retest of first article samples shall not be made until the manufacturer has made necessary corrections in the processing of the material to the satisfaction of the procuring activity. If one of the retest specimens fails the lot shall be permanently rejected with no further testing permitted.

4.8.2 Ballistic. Rejection and retest of ballistic test plates shall be in accordance with A.5.2.

4.8.3 Stress corrosion. Immediately upon notification of the failure of the stress corrosion specimens, the manufacturer may at their expense test thirteen additional specimens for retesting. A minimum of seven specimens out of the thirteen additional specimens shall show no cracks upon examination at the end of 96 hours when tested as specified in 4.7.5. If the retest also fails, the manufacturer may elect to resubmit the lot after retreatment of the entire lot. After retreatment, the lot must pass the requirements outlined in 4.7.2, 4.7.4 and 4.7.5.

4.9 Reduced testing. At the discretion of the procuring activity and as specified in the contract or purchase order (see 6.2), the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product meeting the testing requirements is being produced and providing the manufacturer agrees to maintain the same manufacturing procedures. Testing for a given plate thickness shall return to standard (non-reduced testing) conditions of one plate per lot, whenever a ballistic test plate fails to meet ballistic requirements.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When packaging of materiel components are to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The armor specified herein is intended for use on combat and tactical vehicles to protect the occupants against small arms fire, fragments, and shrapnel. Pratt & Miller Engineering who has been developing the OCP-TECD CAMEL demonstrator plans to use a weldable 2139 aluminum alloy.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Specify classification of alloy, and ordered thickness (see 1.1 and 1.3)
(c) When first article is required (see 3.1 and 4.3).
(d) Special mechanical properties and ballistic requirements, if required (see 3.3).
(e) When stress corrosion testing is required and if the test conditions are different (see 3.6, 4.5.2.4, and 4.7.5).
(f) Dimension and tolerance requirements if different (see 3.7 and 3.7.1).
(g) If markings are different and when impression stamping is permissible (see 3.8).
(h) Define surface cracks, edge cracks and edge laminations for the specific application (see 3.10.1).
(i) If the weight of finished plate can exceed 50,000 pounds (see 4.2).
(j) The orientation of the ballistic plate with respect to rolling is different (see 4.5.1.3 and 4.5.2.3).
(k) Dimensional sampling procedure approved by the procuring activity (see 4.6.2).
(l) If approval was requested and received for a different ballistic testing facility (see 4.7.3.1).
(m) Rejection and retest requirement, if different (see 4.8).
(n) If reduced testing is allowed (see 4.9).
(o) Packaging requirements (see 5.1).
(p) Striking velocities for those thicknesses not covered (see A.3.6.1.1).

6.3 Weldability. The Army Research Lab has conducted testing and found that class I, type A of AA2139 can be welded and demonstrates good shock resistance when tested with the method described in MIL-STD-3057, "Arc Welding of Armor Grade Aluminum". However, MIL-STD-

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3057 does not currently address this alloy in thicknesses other than 0.5 and 1.5 inches. Qualification of new plate thicknesses and configurations under PPIT may be achieved in accordance with MIL-STD-3057, section 5.3.6.3.9: Proofing projectile and velocity. Tests have demonstrated that material covered by this specification is weldable to itself and other weldable 2000 series alloys using a 2000 or 4000 series filler and is weldable to 6000 series alloys using a 4000 series filler. It is not weldable to 5000 or 7000 series alloys by conventional welding techniques. Any deviations from these conditions should be demonstrated in accordance with MIL-STD-3057 prior to acceptance by the procuring activity.

6.4 Weld wire. Corner joints were successfully welded with AA4043 weld wire for class I, type A (AA2139) material when tested with the required proofing projectile per MIL-STD-3057 (see 1.2).

6.5 Density. The density of AA2139 is 0.102 lb/in^3 (2.81 g/cm^3). The density of AA2195 is 0.098 lb/in^3 (2.71 g/cm^3). The density of AA2060 is 0.098lb/in^3 (2.72 g/cm^3).

6.6 Tempers. The following tempers have shown to produce acceptable material when used to obtain the desired condition; maximum resistance to armor piercing (AP) projectiles, or maximum resistance to blast.

For class I, type A = T8
For class I, type B = T84

6.7 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

6.8 Subject term (key word) listing.

Ballistic testing	Caliber .50 FSP	20-mm FSP
Caliber .30 AP M2	Military vehicles	20-mm API-T M602
Caliber .50 AP M2	Stress corrosion	14.5-mm API

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6.7 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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APPENDIX A

ARMOR PLATE, ALUMINUM, ALLOY 2139 WELDABLE & ALLOY 2195 and 2060 UNWELDABLE APPLIQUE

A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor plate, AA2139, AA2195, and AA2060 when tested in accordance with the provisions of this specification. When there is mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification and the information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662 - V50 Ballistic Test for Armor

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

A.3 DEFINITIONS

A.3.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

A.3.2 Fair impact. An impact may be considered fair when an un-yawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity at a distance of at least two projectile diameters from any previous impact or disturbed area resulting from an impact, or from any crack, or from any edge of the test specimen.

A.3.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

A.3.4 Partial penetration, (PP). Any fair impact which is not a complete penetration may be considered a partial penetration.

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A.3.5 Witness plate. A thin sheet located behind and parallel to the ballistic test sample which is used to detect penetrating projectiles or spall.

A.4 REQUIREMENTS

A.4.1 Resistance to penetration. The minimum required V50 ballistic limit shall be in accordance with the values shown in tables A-I through A-VII.

A.5 TESTS

A.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V50 Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A.5.1.1 Temperature Conditioning. Prior to the test, the test item(s) shall be temperature conditioned at least eight hours. Thermostatic control shall be such that the average temperature of the item during the test shall be 72 ± 15°F (22 ± 8°C).

A.5.1.2 Protection ballistic limit, BL(P).

A.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) shall be reported.

A.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data shall be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit shall be computed on the basis of 4- or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing shall continue until a 10-round ballistic limit has been attained using the smallest possible velocity spread. Ten-round ballistic limits shall be reported as agreed upon between the contractor and procuring activity.

A.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit, which has been determined, is within ± 10 fps from the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, shall be fired to reduce the gap to 25 fps or less. The ballistic limit shall then be recomputed using the above criteria. The recomputed BL(P) shall be reported as the BL(P) of the plate (in borderline cases, a reduction of the gap between the

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high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P)).

A.5.2 Rejection and retest of ballistic plates.

A.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or purchase order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

A.5.2.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates shall be submitted for first article testing, and both tests shall pass; otherwise, the armor material shall be rejected.

A.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or purchase order, failure of a test plate to meet the ballistic requirements indicates failure of the lot; however, the final decision shall depend on the outcome of retests, if submitted.

A.5.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure may submit at their own expense two additional test plates from the same lot for ballistic retest. If either of these plates fails the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fails, the lot shall be permanently rejected.

A.5.3 Disposal of ballistic test plates.

A.5.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates shall be returned "as is" to the applicant, at their own expense, unless the plates were destroyed in testing.

A.5.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered part of the lot they represent, and ownership of the test plates passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered part of the lot they represent and remain the property of the producer. The now rejected lot also remains the property of the producer. The failed plates shall be returned, upon request, as in A.5.3.1.

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TABLE A-I. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 30° obliquity.

Thickness, inches	Required BL(P), fps			Thickness, inches	Required BL(P), fps		
	Class I	Class II	Class III		Class I	Class II	Class III
0.475	1459	1496	1582	0.635	1799	1831	1911
0.480	1471	1508	1594	0.640	1808	1840	1920
0.485	1482	1519	1605	0.645	1818	1850	1929
0.490	1494	1531	1616	0.650	1827	1859	1939
0.495	1505	1542	1627	0.655	1836	1869	1948
0.500 1/	1516	1553	1638	0.660	1845	1878	1957
0.505	1527	1564	1649	0.665	1854	1887	1966
0.510	1538	1575	1660	0.670	1863	1896	1975
0.515	1549	1586	1670	0.675	1872	1905	1984
0.520	1559	1597	1681	0.680	1881	1915	1993
0.525	1570	1608	1692	0.685	1890	1924	2002
0.530	1581	1619	1702	0.690	1899	1933	2011
0.535	1591	1630	1713	0.695	1908	1942	2020
0.540	1602	1640	1723	0.700	1916	1951	2029
0.545	1612	1651	1734	0.705	1925	1960	2038
0.550	1623	1661	1744	0.710	1934	1969	2047
0.555	1633	1672	1754	0.715	1942	1977	2055
0.560	1643	1682	1764	0.720	1951	1986	2064
0.565	1653	1692	1775	0.725	1960	1995	2073
0.570	1663	1703	1785	0.730	1968	2004	2081
0.575	1673	1713	1795	0.735	1977	2012	2090
0.580	1693	1723	1805	0.740	1985	2021	2099
0.585	1703	1733	1815	0.745	1993	2030	2107
0.590	1713	1743	1824	0.749 2/	2002	2037	2114
0.595	1723	1753	1834	0.755	2010	2047	2124
0.600	1733	1763	1844	0.760	2019	2055	2132
0.605	1742	1773	1854	0.765	2027	2064	2141
0.610	1752	1783	1863	0.770	2035	2072	2149
0.615	1761	1792	1873	0.775	2043	2081	2157
0.620	1771	1802	1882	0.780	2051	2089	2166
0.625	1780	1812	1892	0.785	2060	2097	2174
0.630	1790	1821	1901	0.790	2068	2105	2182

1/ Specification requirements begin for this ordered thickness.
2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-II. Minimum required ballistic limits – caliber .50 fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Required BL(P), fps
	Class I & II	Class III		Class I & II	Class III
0.730	1986	2054	0.860	2601	2650
0.735	2007	2074	0.865	2628	2676
0.740	2028	2095	0.870	2656	2703
0.745	2049	2115	0.875	2683	2729
0.750 [1/]	2070	2136	0.880	2711	2756
0.755	2092	2157	0.885	2739	2783
0.760	2114	2178	0.890	2768	2811
0.765	2136	2200	0.895	2797	2838
0.770	2158	2222	0.900	2826	2866
0.775	2181	2243	0.905	2856	2894
0.780	2203	2266	0.910	2885	2923
0.785	2226	2288	0.915	2915	2952
0.790	2250	2310	0.920	2946	2981
0.795	2273	2333	0.925	2976	3010
0.800	2297	2356	0.930	3007	3040
0.805	2321	2379	0.935	3039	3070
0.810	2345	2403	0.940	3071	3100
0.815	2369	2426	0.945	3103	3131
0.820	2394	2450	0.950 [2/]	3135	3161
0.825	2419	2474	0.955	3168	3192
0.830	2444	2499	0.960	3201	3224
0.835	2470	2523	0.965	3234	3256
0.840	2495	2548	0.970	3268	3288
0.845	2521	2573	0.975	3302	3320
0.850	2548	2599	0.980	3336	3353
0.855	2574	2624	0.985	3371	3386

[1/] Specification requirements begin for this ordered thickness.
[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-III. Minimum required ballistic limits – 20mm fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps Classes I, II & III	Thickness, inches	Required BL(P), fps Classes I, II & III
0.845	1139	0.975	1362
0.850	1147	0.980	1372
0.855	1155	0.985	1381
0.860	1162	0.990	1391
0.865	1171	0.995	1401
0.870	1179	1.000	1412
0.875	1187	1.005	1420
0.880	1195	1.010	1430
0.885	1203	1.015	1440
0.890	1212	1.020	1450
0.895	1220	1.025	1460
0.900	1228	1.030	1470
0.905	1237	1.035	1480
0.910	1246	1.040	1490
0.915	1254	1.045	1501
0.920	1263	1.050	1511
0.925	1272	1.055	1522
0.930	1280	1.060	1532
0.935	1289	1.065	1543
0.940	1298	1.070	1553
0.945	1307	1.075	1564
0.951 1/	1318	1.080	1575
0.955	1325	1.085	1586
0.960	1335	1.090	1597
0.965	1344	1.095	1608
0.970	1353	1.100	1619

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-III. Minimum required ballistic limits – 20mm fragment simulating projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps Classes I, II & III	Thickness, inches	Required BL(P), fps Classes I, II & III
1.105	1630	1.230	1937
1.110	1642	1.235	1951
1.115	1653	1.240	1964
1.120	1664	1.245	1978
1.125	1676	1.250	1991
1.130	1687	1.255	2005
1.135	1699	1.260	2019
1.140	1711	1.265	2033
1.145	1723	1.270	2047
1.150	1735	1.275	2061
1.155	1747	1.280	2076
1.160	1759	1.285	2090
1.165	1771	1.290	2104
1.170	1783	1.295	2119
1.175	1796	1.300	2134
1.180	1808	1.305	2148
1.185	1821	1.310	2163
1.190	1833	1.315	2178
1.195	1846	1.320	2193
1.200	1859	1.325	2209
1.205	1872	1.330	2224
1.210	1884	1.335	2239
1.215	1898	1.340	2255
1.220	1911	1.345	2270
1.225	1924	1.350	2286

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-III. Minimum required ballistic limits – 20mm fragment simulating projectiles at 0° obliquity (continued).

Thickness, inches	**Required BL(P), fps** **Classes I, II & III**	**Thickness, inches**	**Required BL(P), fps** **Classes I, II & III**
1.355	2302	1.485	2754
1.360	2318	1.490	2773
1.365	2334	1.495	2793
1.370	2350	1.500 2/	2812
1.375	2366	1.505	2831
1.380	2383	1.510	2851
1.385	2399	1.515	2871
1.390	2416	1.520	2891
1.395	2433	1.525	2911
1.400	2449	1.530	2931
1.405	2466	1.535	2951
1.410	2484	1.540	2972
1.415	2501	1.545	2992
1.420	2518	1.550	3013
1.425	2535	1.555	3034
1.430	2553	1.560	3055
1.435	2571	1.565	3076
1.440	2589	1.570	3097
1.445	2606	1.575	3119
1.450	2624	1.580	3140
1.455	2643	1.585	3162
1.460	2661	1.590	3184
1.465	2679	1.595	3206
1.470	2698	1.600	3228
1.475	2717	1.605	3251
1.480	2735	1.610	3273

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Required BL(P), fps
	Class I & II	Class III		Class I & II	Class III
0.730	1799	1828	0.905	2059	2112
0.735	1807	1837	0.910	2066	2120
0.740	1815	1846	0.915	2073	2127
0.745	1823	1854	0.920	2080	2135
0.750 1/	1831	1863	0.925	2086	2142
0.755	1839	1871	0.930	2093	2150
0.760	1846	1880	0.935	2100	2157
0.765	1854	1888	0.940	2107	2165
0.770	1862	1897	0.945	2114	2172
0.775	1870	1905	0.950	2120	2179
0.780	1877	1914	0.955	2127	2187
0.785	1885	1922	0.960	2134	2194
0.790	1892	1930	0.965	2141	2201
0.795	1900	1939	0.970	2147	2209
0.800	1907	1947	0.975	2154	2216
0.805	1915	1955	0.980	2161	2223
0.810	1922	1963	0.985	2167	2230
0.815	1930	1971	0.990	2174	2237
0.820	1937	1979	0.995	2180	2245
0.825	1945	1987	1.000	2187	2252
0.830	1952	1995	1.005	2193	2259
0.835	1959	2003	1.010	2200	2266
0.840	1966	2011	1.015	2206	2273
0.845	1974	2019	1.020	2213	2280
0.850	1981	2027	1.025	2219	2287
0.855	1988	2035	1.030	2226	2294
0.860	1995	2043	1.035	2232	2301
0.865	2003	2051	1.040	2239	2308
0.870	2010	2059	1.045	2245	2315
0.875	2017	2066	1.050	2251	2322
0.880	2024	2074	1.055	2258	2329
0.885	2031	2082	1.060	2264	2335
0.890	2038	2089	1.065	2270	2342
0.895	2045	2097	1.070	2277	2349
0.900	2052	2105	1.075	2283	2356

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps Class I & II	Required BL(P), fps Class III	Thickness, inches	Required BL(P), fps Class I & II	Required BL(P), fps Class III
1.080	2289	2363	1.255	2498	2589
1.085	2295	2369	1.260	2504	2595
1.090	2302	2376	1.265	2510	2601
1.095	2308	2383	1.270	2515	2607
1.100	2314	2390	1.275	2521	2613
1.105	2320	2396	1.280	2527	2620
1.110	2326	2403	1.285	2532	2626
1.115	2332	2410	1.290	2538	2632
1.120	2339	2416	1.295	2544	2638
1.125	2345	2423	1.300	2549	2644
1.130	2351	2429	1.305	2555	2650
1.135	2357	2436	1.310	2561	2656
1.140	2363	2443	1.315	2566	2662
1.145	2369	2449	1.320	2572	2668
1.150	2375	2456	1.325	2577	2674
1.155	2381	2462	1.330	2583	2680
1.160	2387	2469	1.335	2588	2686
1.165	2393	2475	1.340	2594	2692
1.170	2399	2482	1.345	2599	2698
1.175	2405	2488	1.350	2605	2704
1.180	2411	2494	1.355	2610	2710
1.185	2417	2501	1.360	2616	2716
1.190	2423	2507	1.365	2621	2721
1.195	2429	2514	1.370	2627	2727
1.200	2435	2520	1.375	2632	2733
1.205	2440	2526	1.335	2588	2686
1.210	2446	2533	1.340	2594	2692
1.215	2452	2539	1.345	2599	2698
1.220	2458	2545	1.350	2605	2704
1.225	2464	2551	1.355	2610	2710
1.230	2470	2558	1.360	2616	2716
1.235	2475	2564	1.365	2621	2721
1.240	2481	2570	1.370	2627	2727
1.245	2487	2576	1.375	2632	2733
1.250	2493	2583	1.380	2638	2739

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps Class I & II	Required BL(P), fps Class III	Thickness, inches	Required BL(P), fps Class I & II	Required BL(P), fps Class III
1.385	2643	2745	1.455	2718	2825
1.390	2648	2751	1.460	2723	2831
1.395	2654	2756	1.465	2728	2837
1.400	2659	2762	1.470	2733	2842
1.405	2665	2768	1.475	2739	2848
1.410	2670	2774	1.480	2744	2853
1.415	2675	2780	1.485	2749	2859
1.420	2681	2785	1.490	2754	2865
1.425	2686	2791	1.495	2759	2870
1.430	2691	2797	1.500 2/	2765	2876
1.435	2697	2802	1.505	2770	2881
1.440	2702	2808	1.510	2775	2887
1.445	2707	2814	1.515	2780	2892
1.450	2712	2820	1.520	2785	2898

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps			Thickness, inches	Required BL(P), fps		
	Class I	Class II	Class III		Class I	Class II	Class III
1.480	2070	2083	2066	1.630	2193	2210	2193
1.485	2074	2087	2070	1.635	2197	2214	2197
1.490	2079	2092	2075	1.640	2201	2219	2201
1.495	2083	2096	2079	1.645	2205	2223	2205
1.501 1,2/	2088	2100	2084	1.650	2209	2227	2209
1.505	2091	2105	2088	1.655	2213	2231	2213
1.510	2095	2109	2092	1.660	2217	2235	2217
1.515	2100	2113	2096	1.665	2221	2239	2221
1.520	2104	2118	2100	1.670	2225	2243	2225
1.525	2108	2122	2105	1.675	2229	2247	2229
1.530	2112	2126	2109	1.680	2233	2251	2233
1.535	2116	2131	2113	1.685	2237	2255	2237
1.540	2120	2135	2118	1.690	2241	2259	2241
1.545	2124	2139	2122	1.695	2245	2263	2245
1.550	2129	2143	2126	1.700	2248	2267	2249
1.555	2133	2148	2130	1.705	2252	2271	2253
1.560	2137	2152	2134	1.710	2256	2275	2257
1.565	2141	2156	2139	1.715	2260	2279	2261
1.570	2145	2160	2143	1.720	2264	2283	2265
1.575	2149	2165	2147	1.725	2268	2287	2269
1.580	2153	2169	2151	1.705	2252	2271	2253
1.585	2157	2173	2155	1.710	2256	2275	2257
1.590	2161	2177	2160	1.715	2260	2279	2261
1.595	2165	2181	2164	1.720	2264	2283	2265
1.600	2169	2185	2168	1.725	2268	2287	2269
1.605	2173	2190	2172	1.730	2272	2291	2273
1.610	2177	2194	2176	1.735	2276	2295	2277
1.615	2181	2198	2180	1.740	2279	2299	2281
1.620	2185	2202	2184	1.745	2283	2303	2285
1.625	2189	2206	2189	1.750	2287	2307	2289

1/ Specification requirements begin for this ordered thickness.
2/ Class I, type B minimum required ballistic limits shall be -90fps vs. listed thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps			Thickness, inches	Required BL(P), fps		
	Class I	Class II	Class III		Class I	Class II	Class III
1.755	2291	2311	2293	1.880	2384	2408	2389
1.760	2295	2315	2297	1.885	2388	2412	2393
1.765	2298	2319	2301	1.890	2392	2415	2397
1.770	2302	2323	2305	1.895	2395	2419	2400
1.775	2306	2327	2309	1.900	2399	2423	2404
1.780	2310	2331	2313	1.905	2403	2427	2408
1.785	2314	2335	2316	1.910	2406	2431	2412
1.790	2317	2339	2320	1.915	2410	2434	2415
1.795	2321	2343	2324	1.920	2414	2438	2419
1.800	2325	2346	2328	1.925	2417	2442	2423
1.805	2329	2350	2332	1.930	2421	2446	2427
1.810	2332	2354	2336	1.935	2424	2449	2430
1.815	2336	2358	2340	1.940	2428	2453	2434
1.820	2340	2362	2344	1.945	2432	2457	2438
1.825	2344	2366	2347	1.950	2435	2460	2441
1.830	2347	2370	2351	1.955	2439	2464	2445
1.835	2351	2374	2355	1.960	2442	2468	2449
1.840	2355	2377	2359	1.965	2446	2472	2452
1.845	2359	2381	2363	1.970	2450	2475	2456
1.850	2362	2385	2366	1.975	2453	2479	2460
1.830	2347	2370	2351	1.980	2457	2483	2463
1.835	2351	2374	2355	1.985	2460	2486	2467
1.840	2355	2377	2359	1.990	2464	2490	2471
1.845	2359	2381	2363	1.995	2467	2494	2474
1.850	2362	2385	2366	2.000	2471	2497	2478
1.855	2366	2389	2370	2.005	2474	2501	2482
1.860	2370	2393	2374	2.010	2478	2505	2485
1.865	2373	2397	2378	2.015	2482	2508	2489
1.870	2377	2400	2382	2.020	2485	2512	2492
1.875	2381	2404	2385	2.025	2489	2515	2496

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps			Thickness, inches	Required BL(P), fps		
	Class I	Class II	Class III		Class I	Class II	Class III
2.030	2492	2519	2500	2.205	2612	2643	2623
2.035	2496	2523	2503	2.210	2615	2646	2626
2.040	2499	2526	2507	2.215	2619	2650	2629
2.045	2503	2530	2510	2.220	2622	2653	2633
2.050	2506	2534	2514	2.225	2625	2657	2636
2.055	2510	2537	2518	2.230	2629	2660	2640
2.060	2513	2541	2521	2.235	2632	2664	2643
2.065	2517	2544	2525	2.240	2635	2667	2647
2.070	2520	2548	2528	2.245	2639	2671	2650
2.075	2523	2552	2532	2.250 [3/]	2642	2674	2653
2.080	2527	2555	2535	2.255	2645	2678	2657
2.085	2530	2559	2539	2.260	2649	2681	2660
2.090	2534	2562	2542	2.265	2652	2685	2663
2.095	2537	2566	2546	2.270	2655	2688	2667
2.100	2541	2569	2550	2.275	2659	2694	2670
2.105	2544	2573	2553	2.280	2662	**N/A**	2674
2.110	2548	2576	2557	2.285	2665	**N/A**	2677
2.115	2551	2580	2560	2.290	2668	**N/A**	2680
2.120	2554	2584	2564	2.295	2672	**N/A**	2684
2.125	2558	2587	2567	2.300	2675	**N/A**	2687
2.130	2561	2591	2571	2.305	2678	**N/A**	2690
2.135	2565	2594	2574	2.310	2681	**N/A**	2694
2.140	2568	2598	2578	2.315	2685	**N/A**	2697
2.145	2572	2601	2581	2.320	2688	**N/A**	2700
2.150	2575	2605	2585	2.325	2691	**N/A**	2704
2.180	2595	2626	2605	2.330	2694	**N/A**	2707
2.185	2599	2629	2609	2.335	2698	**N/A**	2710
2.190	2602	2633	2612	2.340	2701	**N/A**	2714
2.195	2605	2636	2616	2.345	2704	**N/A**	2717
2.200	2609	2640	2619	2.350	2707	**N/A**	2720

[3] Specification requirements end for this ordered thickness for class II.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps			Thickness, inches	Required BL(P), fps		
	Class I	Class II	Class III		Class I	Class II	Class III
2.355	2711	N/A	2724	2.495	2800	N/A	2815
2.360	2714	N/A	2727	2.500 [4/]	2803	N/A	2818
2.365	2717	N/A	2730	2.505	2806	N/A	2821
2.370	2720	N/A	2733	2.510	2809	N/A	2824
2.375	2724	N/A	2737	2.515	2812	N/A	2827
2.380	2727	N/A	2740	2.520	2815	N/A	2831
2.385	2730	N/A	2743	2.525	2818	N/A	2834
2.390	2733	N/A	2747	2.530	2821	N/A	N/A
2.395	2736	N/A	2750	2.535	2824	N/A	N/A
2.400	2740	N/A	2753	2.540	2828	N/A	N/A
2.405	2743	N/A	2756	2.545	2831	N/A	N/A
2.410	2746	N/A	2760	2.550	2834	N/A	N/A
2.415	2749	N/A	2763	2.555	2837	N/A	N/A
2.420	2752	N/A	2766	2.560	2840	N/A	N/A
2.425	2755	N/A	2769	2.565	2843	N/A	N/A
2.430	2759	N/A	2773	2.570	2846	N/A	N/A
2.435	2762	N/A	2776	2.575	2849	N/A	N/A
2.440	2765	N/A	2779	2.580	2852	N/A	N/A
2.445	2768	N/A	2782	2.585	2855	N/A	N/A
2.450	2771	N/A	2786	2.590	2858	N/A	N/A
2.455	2774	N/A	2789	2.595	2861	N/A	N/A
2.460	2778	N/A	2792	2.600 [5/]	2864	N/A	N/A
2.465	2781	N/A	2795	2.605	2867	N/A	N/A
2.470	2784	N/A	2799	2.610	2871	N/A	N/A
2.475	2787	N/A	2802	2.615	2874	N/A	N/A
2.480	2790	N/A	2805	2.620	2877	N/A	N/A
2.485	2793	N/A	2808	2.625	2880	N/A	N/A
2.490	2796	N/A	2811	2.630	N/A	N/A	N/A

[4/] Specification requirements end for this ordered thickness for class I, type A & Class III.
[5/] Specification requirements end for this ordered thickness for class I, type B.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-VI. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps Class I	Thickness, inches	Required BL(P), fps Class I	Thickness, inches	Required BL(P), fps Class I
2.480	2577	2.655	2689	2.830	2797
2.485	2580	2.660	2692	2.835	2800
2.490	2583	2.665	2695	2.840	2803
2.495	2586	2.670	2698	2.845	2806
2.501 1/	2590	2.675	2702	2.850	2809
2.505	2593	2.680	2705	2.855	2812
2.510	2596	2.685	2708	2.860	2815
2.515	2600	2.690	2711	2.865	2818
2.520	2603	2.695	2714	2.870	2821
2.525	2606	2.700	2717	2.875	2824
2.530	2609	2.705	2720	2.880	2827
2.535	2613	2.710	2723	2.885	2830
2.540	2616	2.715	2727	2.890	2833
2.545	2619	2.720	2730	2.895	2836
2.550	2622	2.725	2733	2.900	2839
2.555	2625	2.730	2736	2.905	2842
2.560	2629	2.735	2739	2.910	2845
2.565	2632	2.740	2742	2.915	2848
2.570	2635	2.745	2745	2.920	2851
2.575	2638	2.750	2748	2.925	2854
2.580	2641	2.755	2751	2.930	2857
2.585	2645	2.760	2754	2.935	2860
2.590	2648	2.765	2757	2.940	2863
2.595	2651	2.770	2760	2.945	2866
2.600	2654	2.775	2763	2.950	2869
2.605	2657	2.780	2767	2.955	2872
2.610	2661	2.785	2770	2.960	2874
2.615	2664	2.790	2773	2.965	2877
2.620	2667	2.795	2776	2.970	2880
2.625	2670	2.800	2779	2.975	2883
2.630	2673	2.805	2782	2.980	2886
2.635	2676	2.810	2785	2.985	2889
2.640	2680	2.815	2788	2.990	2892
2.645	2683	2.820	2791	2.995	2895
2.650	2686	2.825	2794	3.000	2898

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-VI. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps Class I	Thickness, inches	Required BL(P), fps Class I	Thickness, inches	Required BL(P), fps Class I
3.005	2901	3.180	3001	3.355	3098
3.010	2904	3.185	3004	3.360	3101
3.015	2907	3.190	3007	3.365	3104
3.020	2910	3.195	3010	3.370	3106
3.025	2912	3.200	3012	3.375	3109
3.030	2915	3.205	3015	3.380	3112
3.035	2918	3.210	3018	3.385	3114
3.040	2921	3.215	3021	3.390	3117
3.045	2924	3.220	3024	3.395	3120
3.050	2927	3.225	3026	3.400	3123
3.055	2930	3.230	3029	3.405	3125
3.060	2933	3.235	3032	3.410	3128
3.065	2936	3.240	3035	3.415	3131
3.070	2938	3.245	3037	3.420	3133
3.075	2941	3.250	3040	3.425	3136
3.080	2944	3.255	3043	3.430	3139
3.085	2947	3.260	3046	3.435	3141
3.090	2950	3.265	3049	3.440	3144
3.095	2953	3.270	3051	3.445	3147
3.100	2956	3.275	3054	3.450	3150
3.105	2959	3.280	3057	3.455	3152
3.110	2961	3.285	3060	3.460	3155
3.115	2964	3.290	3062	3.465	3158
3.120	2967	3.295	3065	3.470	3160
3.125	2970	3.300	3068	3.475	3163
3.130	2973	3.305	3071	3.480	3166
3.135	2976	3.310	3073	3.485	3168
3.140	2978	3.315	3076	3.490	3171
3.145	2981	3.320	3079	3.495	3174
3.150	2984	3.325	3082	3.500 2/	3176
3.155	2987	3.330	3084	3.505	3179
3.160	2990	3.335	3087	3.510	3182
3.165	2993	3.340	3090	3.515	3184
3.170	2995	3.345	3093	3.520	3187
3.175	2998	3.350	3095	3.525	3190

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-VII. Minimum required ballistic limits - 20mm M602 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps Class I	Thickness, inches	Required BL(P), fps Class I	Thickness, inches	Required BL(P), fps Class I
3.480	TBD	3.665	TBD	3.850	TBD
3.485	TBD	3.670	TBD	3.855	TBD
3.490	TBD	3.675	TBD	3.860	TBD
3.495	TBD	3.680	TBD	3.865	TBD
3.501 1/	TBD	3.685	TBD	3.870	TBD
3.505	TBD	3.690	TBD	3.875	TBD
3.510	TBD	3.695	TBD	3.880	TBD
3.515	TBD	3.700	TBD	3.885	TBD
3.520	TBD	3.705	TBD	3.890	TBD
3.525	TBD	3.710	TBD	3.895	TBD
3.530	TBD	3.715	TBD	3.900	TBD
3.535	TBD	3.720	TBD	3.905	TBD
3.540	TBD	3.725	TBD	3.910	TBD
3.545	TBD	3.730	TBD	3.915	TBD
3.550	TBD	3.735	TBD	3.920	TBD
3.555	TBD	3.740	TBD	3.925	TBD
3.560	TBD	3.745	TBD	3.930	TBD
3.565	TBD	3.750	TBD	3.935	TBD
3.570	TBD	3.755	TBD	3.940	TBD
3.575	TBD	3.760	TBD	3.945	TBD
3.580	TBD	3.765	TBD	3.950	TBD
3.585	TBD	3.770	TBD	3.955	TBD
3.590	TBD	3.775	TBD	3.960	TBD
3.595	TBD	3.780	TBD	3.965	TBD
3.600	TBD	3.785	TBD	3.970	TBD
3.605	TBD	3.790	TBD	3.975	TBD
3.610	TBD	3.795	TBD	3.980	TBD
3.615	TBD	3.800	TBD	3.985	TBD
3.620	TBD	3.805	TBD	3.990	TBD
3.625	TBD	3.810	TBD	3.995	TBD
3.630	TBD	3.815	TBD	4.000 2/	TBD
3.635	TBD	3.820	TBD	4.010	TBD
3.640	TBD	3.825	TBD	4.015	TBD
3.645	TBD	3.830	TBD	4.020	TBD
3.650	TBD	3.835	TBD	4.025	TBD
3.655	TBD	3.840	TBD	4.030	TBD
3.660	TBD	3.845	TBD	4.035	TBD

1/ Specification requirements begin for this ordered thickness.
2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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CONCLUDING MATERIAL

Custodians:
Army - MR

Preparing activity:
ARMY - MR

(Project 9535-2024-002)

Review activities:
Army - AR, AT, AV, TE
DLA - CQ

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